

Steff Horemans

Freelance Data Engineer — Life Science & Pharma

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Languages: Dutch (native), English (fluent)

PROFILE

Freelance data engineer with 4 years of experience and a PhD in the life sciences, specialised in scientific data engineering and infrastructure for the life science and pharma sector. I work end-to-end: from problem definition and business intake to a finished, governed data product, I combine technical breadth (data engineering, BI, AI and operations research) with deep domain knowledge. This lets me extract value from complex scientific data where generic data engineers lack the domain context.

EXPERIENCE

Data Engineer — Atlas Copco | Aug 2022 – Present

First 2 years via Exellys (contracting), then permanent employment

- Worked end-to-end: from business intake and problem definition to a finished data product in production use.
- Built an order-bundling tool that prevents orders from being fragmented across multiple days, structurally reducing the number of required visits; in daily use since delivery.
- Leads a cross-departmental initiative to identify machines containing obsolete parts, with each department mapping the appropriate follow-up actions within its own domain.
- Chaired Communities of Practice on application development and code standards, helping shape company-wide engineering practices.
- Developed operations research models to optimise complex business processes, with experiment tracking via MLFlow.
- Built data pipelines on Databricks using Lakeflow Jobs for orchestration and Databricks Asset Bundles for infrastructure-as-code, supported by CI/CD and Git-based code reviews.
- Acted as technical data product owner: metadata, automated quality checks, access management and data governance/lineage via Unity Catalog.
- Built internal applications using Databricks Apps, Streamlit and Azure (Web Apps, Data Factory, Cosmos DB).

BI Consultant — Modis (now Akkodis) | Jun 2020 – Jul 2022

- Worked directly with a client in the pharma sector: ran intakes and translated reporting needs into concrete dashboards.
- Designed and built Tableau dashboards used structurally in production to support data-driven decision-making.
- Managed databases (MS SQL, Azure Data Factory) and built data pipelines with Alteryx and Tableau Prep to make data reportable.

PhD Researcher, Life Sciences — Université Paris-Saclay | Oct 2015 – Dec 2019

- Independently ran a multi-year research project in molecular genetics, focused on genetic manipulation of bacteria (incl. PCR, qPCR, Gibson Assembly, Lambda Red recombineering, microscopy and plate reading).
- Investigated the link between DNA replication and metabolism; this led to a peer-reviewed publication on moonlighting activities of pyruvate kinase in DNA replication.
- Investigated the influence of genome architecture on the spatial clustering of genes and its effects on gene transcription.
- Presented results at international conferences and mentored an iGEM student team.
- *This background translates directly to my current work: I read scientific literature fluently and understand the molecular mechanisms behind genetic therapies — knowledge needed to model biomedical data correctly, and which a generic data engineer typically lacks.*

PROJECTS

Genetic Disease Data Mesh on Databricks — *personal project (in development)* · github.com/steff-ml

- Reference implementation of a governed data mesh on Databricks linking genetic mutation profiles to therapy and trial eligibility, developed for Duchenne muscular dystrophy.
- The right therapeutic strategy (such as restoring the reading frame via exon skipping or stop-codon readthrough) depends on the specific underlying mutation; modelling this correctly requires deep biological understanding that goes beyond generic data engineering.
- The pattern scales to other conditions where therapy choice depends on the mutation profile, and thus to precision medicine more broadly.
- Demonstrates end-to-end design: medallion architecture, data products with data contracts and quality SLAs, governance via Unity Catalog, and GenAI extraction of trial eligibility from free text.

EDUCATION

- **PhD, Life Sciences** — Université Paris-Saclay — 2015–2019
- **M2 Synthetic Biology** — Université d'Évry-Val-d'Essonne — 2014–2015
- **MSc Biophysics** — KU Leuven — 2012–2014
- **BSc Biochemistry** — KU Leuven — 2009–2012

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Data & platforms:** Databricks (Lakeflow Jobs, Asset Bundles, Unity Catalog, Databricks Apps), Delta Lake, MLFlow
- **Cloud:** Azure (Web Apps, Data Factory, Cosmos DB, Monitoring)
- **BI & visualisation:** Tableau, Tableau Prep, Alteryx
- **Databases:** MS SQL
- **Engineering & other:** Streamlit, CI/CD, Git, operations research modelling

CERTIFICATIONS

- **Postgraduate Applied AI** — Erasmushogeschool Brussel — 2024–2025

PUBLICATIONS

- *Glycolytic pyruvate kinase moonlighting activities in DNA replication initiation and elongation* — BMC Biol.